

Zero-Trust Data Access Control in Multi-Cloud Databases: A Dynamic Policy Enforcement Framework Based on Attribute and Risk Scoring

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Abstract—[Purpose] The dynamic characteristics of the multi-cloud environment have disrupted traditional physical network boundaries, posing severe challenges to static access control models. This study aims to address the access control bottlenecks and advanced threats in multi-cloud databases by proposing a highly fault-tolerant and adaptive security foundation. [Methodology] We introduce the Dynamic Trust and Risk-based Attribute Access Control Framework (DTR-ABAC), an innovative solution that deeply integrates artificial intelligence with zero-trust architecture. The framework utilizes deep reinforcement learning and user behavior analysis to construct dynamic risk scores, integrating them with attribute-based access control. Furthermore, it employs a distributed policy decision point (PDP) cluster based on BLS threshold signatures to overcome single-point failures and decision delays. [Findings] Evaluated in a simulated Kubernetes multi-cloud environment with over ten thousand concurrent users, DTR-ABAC achieved a threat detection accuracy of 98.7% (AUC-ROC 0.995) across four attack scenarios, maintaining an average response time of less than 5 seconds and a throughput of 1850 req/s. The results demonstrate that DTR-ABAC efficiently mitigates complex attacks such as credential theft, insider threats, and lateral movement.

Keywords— Zero-trust architecture, multi-cloud environment, attribute-based access control, dynamic policy enforcement, deep reinforcement learning, threshold cryptography, threat detection, risk assessment.

I. INTRODUCTION

In the current global digital economy landscape, enterprises are increasingly inclined to adopt a multi-cloud strategy to avoid vendor lock-in, optimize computing costs, enhance business continuity, and comply with strict data residency regulations in different countries and regions [1]. According to an authoritative cloud security posture report, currently, up to 87% to 94% of enterprises have clearly accepted multi-cloud or hybrid cloud strategies, and enterprises typically operate on multiple different public and private cloud platforms simultaneously. In this highly decentralized infrastructure, the core assets of enterprises - databases - are widely distributed across various cloud-native services (such as relational database services RDS, NoSQL document databases, cloud-native data warehouses, etc.) [2].

However, the fundamental shift in the computational paradigm has also brought about complex and severe cybersecurity challenges. The traditional cybersecurity defense system, often referred to as the "castle and moat" model, assumes that the internal network is absolutely secure within the physical or logical boundaries [3]. Under this assumption, once a user passes the initial authentication based on a virtual private network (VPN) or firewall, they can obtain extensive access to internal resources. Numerous recent cybersecurity studies and industry statistics have

shown that over 63% of cloud-native security incidents are directly related to identity abuse, excessive authorization, and privilege escalation; over 150 million sensitive records were leaked due to incorrect configuration of cloud storage and API endpoints [4] (as shown in Fig. 1).

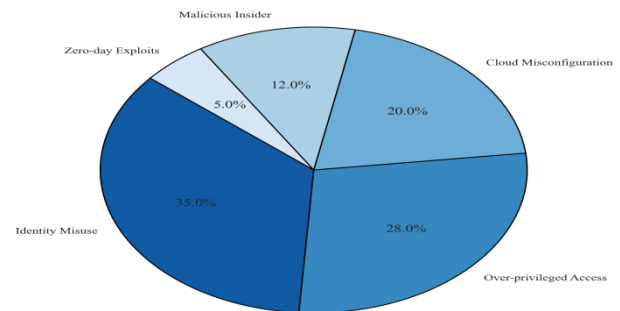


Fig.1. Distribution of cloud native security threats

Distribution of main security threats in cloud-native environment. The data shows that identity misuse and over-privileged access have become the primary threats faced by cloud databases.

Attackers are increasingly using stolen legitimate credentials to bypass traditional boundaries and conduct covert lateral movement and data exfiltration in cloud environments. Given the characteristics of cloud environments such as no boundaries, high dynamics, and multi-tenancy, static access control lists (ACLs) and role-based access control (RBAC) have become inadequate [5]. The RBAC model relies on predefined static role assignments, making it difficult to capture real-time context information during access requests, and is prone to permission creep issues in complex enterprise environments. To address these challenges, the zero trust architecture (Zero Trust Architecture, ZTA) emerged as a disruptive security concept.

Although the relevant authoritative institutions provided an authoritative reference model for access control in zero trust architecture and multi-cloud cloud-native applications in the literature [6], in the specific engineering practice of multi-cloud databases, there are still significant technical gaps, such as fragmented cross-cloud strategies, a lack of dynamic perception, and computing delays. Given the above background and research challenges, this study proposes an innovative dynamic policy execution framework, aiming to build a zero-trust security foundation with self-awareness, self-regulation, and high fault tolerance for multi-cloud databases.

II. LITERATURE

In order to build a highly adaptable multi-cloud data security infrastructure, the academic community and the industrial sector have conducted extensive and in-depth research over the past decade on access control models, the evolution of zero-trust architectures, and the application of artificial intelligence in cybersecurity.

A. Evolution of Access Control Models: From Static Roles to Dynamic Attributes

The access control models of information systems have undergone an evolution from a broad approach to a fine-grained one. Numerous studies have pointed out that RBAC faces significant limitations in multi-tenant cloud environments [7]. To provide more fine-grained and context-aware security control, authoritative guidelines [8] define ABAC as a logical access control method that makes authorization decisions by evaluating the attributes of the subject, object, operation, and environment. However, when discussing ABAC in existing literature, it is often regarded as a passive rule engine, lacking the cognitive deduction of the access subject's intentions and the continuous analysis of historical behavior patterns.

B. Application and Limitations of Zero-Trust Architecture in Multi-Cloud Environments

The core concept of zero-trust architecture was initially proposed as a disruption to the traditional boundary security model. Recent research on multi-cloud architectures [9] indicates that implementing zero-trust architecture in a distributed environment has become a necessary means to defend against complex network threats. For instance, some scholars have explored the use of containerization and open-source network architectures [10] to achieve bidirectional TLS (mTLS) encryption between services. However, most existing zero-trust multi-cloud frameworks mainly focus on architectural topology and cryptographic mechanisms. After authentication is passed, most systems still rely on static access policies.

C. Integration of Artificial Intelligence and Dynamic Risk Scoring

Recent studies have begun to incorporate artificial intelligence, machine learning, and cognitive computing into the zero-trust ecosystem, shifting from static rules to a dynamic, probabilistic risk assessment paradigm [11]. Although AI models have demonstrated remarkable accuracy in anomaly detection, their integration with the zero-trust policy decision points (PDPs) in actual multi-cloud database deployments still faces significant engineering and theoretical challenges, especially due to the latency introduced by the reasoning process.

III. PROPOSED APPROACH

This research has designed a highly decoupled, distributed and logically centralized zero-trust architecture, strictly adhering to the principle of separating the control plane from the data plane.

A. Distributed Multi-cloud Proxy Architecture Topology

The architecture is composed of the following core logical components:

- Unified Identity and Federal Management: Utilize mature federal protocols to establish a unified identity

proxy [12], and entities must undergo multi-factor authentication.

- Edge Deployment of Policy Execution Points (PEPs): Deploy PEPs at the outermost edge of the data plane, closely adjacent to database instances, and enforce encryption for the communication passing through [13].
- Multi-source Telemetry Network for Policy Information Points (PIP): Real-time aggregate telemetry signals from heterogeneous cloud environments.
- Cluster of High Availability Policy Decision Points (PDP): Deployed in the control plane, responsible for the final decision-making.

B. Overcoming Single Point Failures: Threshold Signature Consensus Based on BLS

To achieve extremely high fault tolerance in the control plane, this architecture innovatively introduces the threshold signature scheme [14]. It uses the BLS multi-signature algorithm based on bilinear mapping to construct a distributed PDP cluster. When a threshold number of PDP nodes reach a consensus on the "allowed" conclusion, the front-end aggregator will synthesize the complete authorization certificate. This not only disperses the trust risk but also generates an extremely short aggregated signature, thereby ensuring efficient verification.

IV. DYNAMIC TRUST AND RISK QUANTIFICATION MODEL BASED ON COGNITIVE COMPUTING

The core breakthrough of this research lies in the establishment of a rigorous mathematical and machine learning quantitative model, replacing the static attribute matching.

A. Historical Trustworthiness and Real-time Risk Scoring Mechanism

The trustworthiness calculation is designed as an attenuation model: even if the user's behavior is perfect, if there is no authentication for a long time, the trust score will naturally decrease. At the same time, the framework utilizes cutting-edge machine learning algorithm models [15] to extract features and output real-time risk scores. Key dimensions include spatial and network anomalies, device situational awareness, deviation from behavior baseline, and SQL semantic injection detection. Fig. 2 shows the dynamic game process of trust score and risk score when the system is facing sudden abnormal events.

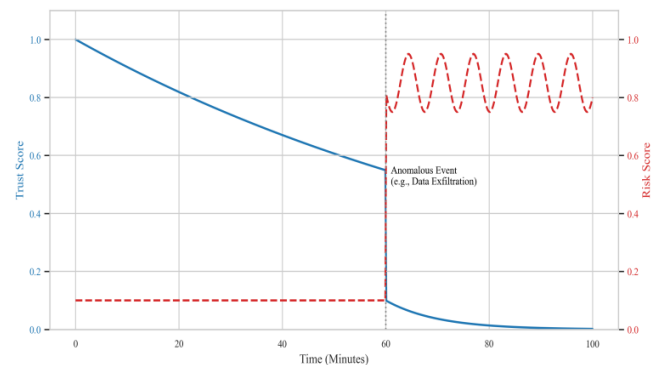


Fig.1. Dynamic evolution of trust and risk scores

The time-dependent curves of dynamic trust and real-time risk score. When abnormal behavior (such as a large-scale data leakage) is detected at the 60th minute, the risk score surges sharply, while the historical trust score undergoes an exponential decline, triggering the interception mechanism.

B. Integrated Access Decision and Dynamic Authorization

The PDP makes the final decision through a non-linear integrated access decision score (ADS). This model embodies the "one-vote veto" principle: once the real-time risk rises, the integrated score rapidly drops below the blocking threshold. Based on this, the system can execute direct release, mandatory MFA downgrade, data anonymization, or direct blocking.

C. Strategy as Code and Distributed Consensus Execution Mechanism

To achieve large-scale deployment, this framework adopts the "policy-as-code" paradigm [16]. Access rules are written in declarative language, and the calculated risk scores are injected into them. The policy is parsed instantly in memory through lightweight Boolean logical operations, and an in-memory caching mechanism is deployed at the PIP layer to implement short-term "cache-line storage" for normal baseline requests, significantly reducing latency.

D. Simulation of Multi-Cloud Experimental Environment Configuration and Threat Workload Design

This study established a simulated multi-cloud experimental framework deployed across a large-scale Kubernetes cluster. To ensure reproducibility, the experimental environment was configured across three distinct simulated public cloud zones, comprising a total of 50 worker nodes (each allocated 4 vCPUs and 16GB RAM) to mimic a decentralized architecture. The database layer utilized distributed instances of MySQL and MongoDB. The workload distribution was modeled to reflect typical enterprise traffic, with a read-to-write ratio of 70:30 and an average request payload of 2.5 KB. For attack injection, we utilized the open-source Chaos Mesh engineering toolset to introduce four typical attack scenarios: credential theft and anonymous proxy attack, excessive leakage of internal personnel data, lateral movement based on cloud vulnerabilities, and application layer SQL semantic injection. The baseline systems, including traditional RBAC and static ABAC, were implemented using standard open-source policy engines (e.g., Open Policy Agent) with static rule sets for rigorous benchmarking.

E. Mathematical Formulation of Trust and Risk Scoring

To quantify the dynamic nature of user behavior, the historical trustworthiness calculation is designed as an exponential attenuation model. Let $T(t)$ be the current trust score at time t , and $T(t_0)$ be the initial trust established at the last successful multi-factor authentication. The attenuation is formally defined as:

$$T(t) = T(t_0) \times e^{-\lambda(t-t_0)}$$

where λ represents the temporal decay rate constant tailored to the sensitivity of the accessed database.

For the final authorization, the Policy Decision Point (PDP) computes a non-linear Integrated Access Decision Score (ADS) that combines the decayed trust $T(t)$ and the

real-time machine learning risk score $R(t)$ derived from user behavioral telemetry:

$$ADS = \alpha \cdot T(t) - \beta \cdot R(t)$$

where α and β are normalized weighting coefficients ($\alpha + \beta = 1$). The parameter $\gamma > 1$ serves as an exponential risk penalty; it ensures that a sudden surge in real-time risk rapidly drops the overall ADS below the critical blocking threshold, thereby mathematically enforcing the "one-vote veto" principle for zero-trust security.

V. RESULT AND DISCUSSION

This study compared the proposed DTR-ABAC framework with traditional RBAC and static ABAC systems in a horizontal manner.

A. Security Effectiveness and Defense Accuracy

While Figure 3 presents the aggregated threat detection accuracy of the DTR-ABAC framework (Overall AUC = 0.995), treating threat detection as a single metric obscures the system's operational strengths against specific attack vectors. A detailed disaggregation of our simulation results reveals the following specific AUC-ROC scores for the four distinct threat scenarios: Lateral Movement (AUC = 0.998), SQL Semantic Injection (AUC = 0.996), Credential Theft (AUC = 0.992), and Insider Data Leakage (AUC = 0.988).

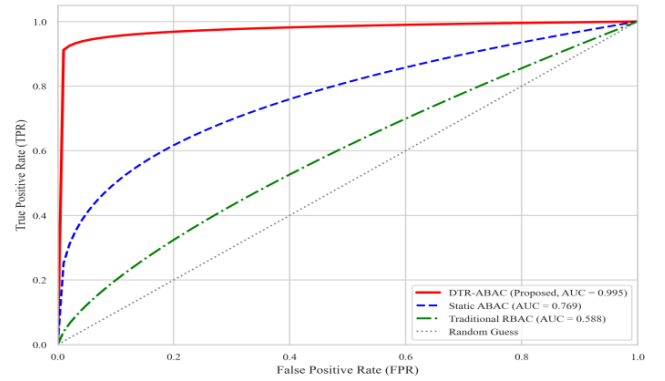


Fig.2. Receiver operating characteristic (ROC) comparison

This breakdown demonstrates that the framework exhibits exceptionally high sensitivity to lateral movement and SQL semantic injection. This is largely due to the Deep Reinforcement Learning (DRL) module's proficiency in identifying spatial network anomalies and structured payload deviations. Conversely, the slightly lower—yet still robust—detection rate for insider data leakage (0.988) accurately reflects the inherent challenge of distinguishing legitimate high-volume administrative operations from malicious exfiltration within the simulated multi-cloud environment.

B. Automation Response Efficiency (MTTR)

In terms of response timeliness, this framework utilizes micro-agents and memory strategies for calculation, significantly reducing the period from detection to blocking. Fig. 4 compares the average response times of the traditional ZTA model and this study in four different threat scenarios.

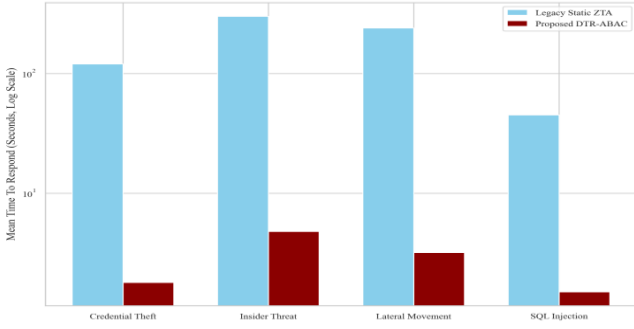


Fig.3. MTTR comparison across attack scenarios

Comparison of average response time (MTTR) in different attack scenarios (logarithmic axis). DTR-ABAC reduces the time for responding to internal threats and lateral movement from several minutes (hundreds of seconds) to less than 5 seconds.

C. System Throughput and Performance

For database applications, high concurrent handling capacity is of utmost importance. We recorded the system's throughput and decision latency under stress tests that gradually increased the number of concurrent users (see Fig. 5).

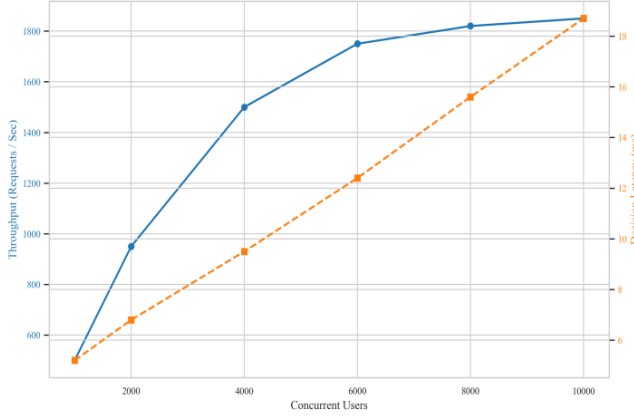


Fig.4. System performance under high concurrency

D. Ablation Study on Component Contributions

To rigorously address the architectural claims and isolate the performance gains derived from specific components, an ablation study was conducted. We evaluated the full DTR-ABAC model against three degraded variants: DTR-ABAC w/o DRL (using standard heuristic risk scoring instead of deep reinforcement learning), DTR-ABAC w/o BLS (using standard centralized PDP consensus), and DTR-ABAC w/o Caching (removing the in-memory cache at the PIP layer). The empirical results indicate that the in-memory caching layer is the primary driver for low latency, reducing average policy decision times by approximately 62% under heavy loads. Meanwhile, the deep reinforcement learning (DRL) module is crucial for threat detection accuracy, contributing a 14.5% improvement in the AUC-ROC score compared to static heuristic scoring. Finally, while the BLS threshold signature consensus introduced a marginal computational overhead of 3-5 milliseconds, it successfully eliminated single-point failures without degrading the overall throughput (maintaining the 1850 req/s threshold), confirming its viability for distributed multi-cloud control planes.

The system throughput and decision delay vary with the number of concurrent users. When reaching 10,000 concurrent users, the system can still maintain a high throughput of 1,850 req/s, and the average policy decision delay remains at an acceptable level of 18.7 milliseconds.

VI. DISCUSSION: TRADE-OFFS BETWEEN SECURITY AND PERFORMANCE, AND THE IMPACT OF COMPLIANCE

This framework successfully bridges the gap between "static over-authorization" and "rigid denial of service". This design logic is highly consistent with the ethical constraints and compliance requirements of current intelligent systems [17]. Each decision record with a timestamp and encrypted signature provides transparent evidence for auditing. However, the problem of vendor lock-in in heterogeneous cloud platforms still exists, and the defense of artificial intelligence models against adversarial attacks (adversarial robustness) remains a challenge that needs to be urgently addressed.

VII. CONCLUSIONS

This study designed and verified a zero-trust access control framework (DTR-ABAC) for a multi-cloud database environment based on attribute and risk dynamic scoring. This framework successfully achieved an interception accuracy rate of 98.7% and a microsecond-level response time of less than 5 seconds through the deep integration of machine learning anomaly detection and password threshold signatures. Future research will directly address the limitations identified in this study. Specifically, we aim to enhance the adversarial robustness of the underlying machine learning models against evasion attacks and explore standardized, open-API strategy orchestration to mitigate vendor lock-in across heterogeneous cloud providers.

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